

LifeMem: Enabling Lifelong Experience Reuse for LLM Agents

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Abstract

Large language model agents are expected to continuously adapt to new tasks and environments over their lifetime by reusing past experience. However, existing memory-based agents struggle to transfer reusable experience across environments and suffer from catastrophic forgetting as experience accumulated. To address these challenges, we propose LifeMem, a life-long learning framework that enables agents to transfer knowledge across multiple environments. During learning, LifeMem clusters accumulated interaction trajectories based on underlying workflows to extract reusable skills. When solving a new task at inference time, the agent recalls relevant skills and trajectories to guide actions. To validate our method, we conduct experiments across 10 environments and over 13k tasks with 2k newly annotated interaction trajectories. Results show that LifeMem enables effective experience reuse in lifelong learning, achieving both reduced forgetting on learned tasks and superior cross-task transfer. Further analysis reveals that task streaming impacts learning, while consolidating structurally similar trajectories within memory boosts performance. Our dataset and code are available at <https://github.com/BITHLP/LifeMem>.

1 Introduction

The pursuit of artificial general intelligence (AGI) requires autonomous agents that can continuously adapt to diverse and evolving environments over their lifetime (Xi et al., 2025; Jiang et al., 2025). Accordingly, large language model (LLM)-based agents are expected to generalize across multiple domains, such as embodied control, web navigation, and tool use, without relying on environment-specific redesigns (Zheng et al., 2025c; Wang et al., 2024). To achieve this, agents have to learn reusable experience from new interaction trajectories without forgetting previously acquired experience (Cheng et al., 2026; Shan et al., 2026),

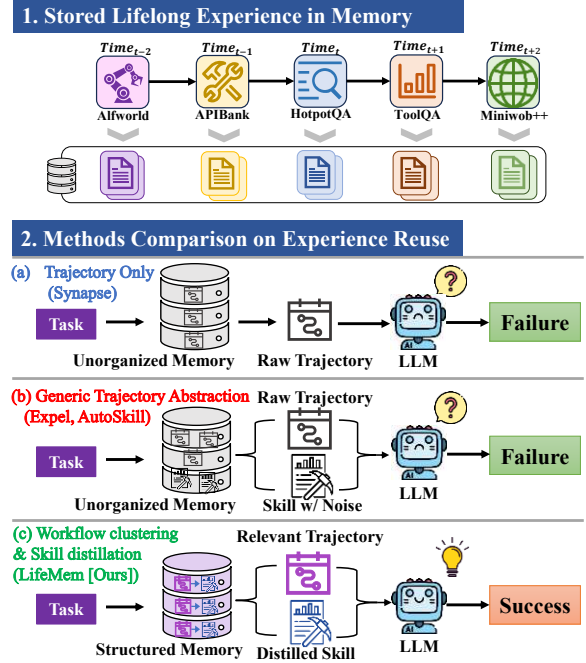


Figure 1: In lifelong learning, agent stores experience in memory. LifeMem enables effective experience reuse compared to previous methods.

a capability known as **lifelong learning** (Zheng et al., 2025a,c). In particular, under the **inter-environment setting**, interaction trajectories are collected from heterogeneous environments with distinct action and observation spaces, which further complicates effective experience reuse (Zheng et al., 2025c).

However, existing LLM-based agents still struggle to effectively reuse past experience to adapt to new environments. *Internal-memory-based methods* (Yao et al., 2023; Qiao and Mahdavi, 2024) fine-tune the model on trajectories from predefined environments, which restricts generalization and incurs high training costs (Tomilin et al., 2025; Zhao et al., 2024a). *External-memory-based methods* maintain an experience database and retrieve relevant trajectories to guide in-context learning (Xi-

ang et al., 2024), yet they encounter challenges as experience accumulates across heterogeneous environments (Zhao et al., 2024b): **(1) trajectory-only retrieval** (Zheng et al., 2024), relying solely on raw trajectories, fails to bridge mismatched action spaces, causing weak generalization. Furthermore, as the database grows, it impairs accurate retrieval, leading to catastrophic forgetting of earlier environments. **(2) generic trajectory abstraction** (Zhao et al., 2024a; Yang et al., 2026), distilling high-level skills directly over an unorganized memory pool, allows irrelevant trajectories to introduce severe cross-environment noise. These noisy and unrepresentative skills are too coarse to guide novel tasks, leading to performance degradation (Figure 1).

To address these challenges, we propose **LifeMem**, a lifelong learning framework that enables agents to accumulate and reuse experience across multiple environments. LifeMem comprises two key components: **(1) workflow clustering**, which partitions trajectories based on underlying action workflows rather than surface semantic similarity, avoiding cross-environment retrieval interference. **(2) skill distillation**, instead of global abstraction, it distills skills locally from each workflow cluster, yielding highly representative skills untainted by noise. During **inference**, the agent leverages both environment-aligned trajectories and associated skills to support effective experience reuse without action-space interference.

To validate the effectiveness of our method, we conduct experiments on 5 widely used agent scenarios (Chang et al., 2024; Liu et al., 2023b): embodied action, tool utilization, web browsing, web search, and data analysis, covering 10 environments and more than 13k tasks with interaction trajectories (Yao et al., 2022b). Notably, for 4 environments that lack interaction trajectories in existing benchmarks, we additionally annotate over 2k trajectories as training data. Experiment results show that LifeMem demonstrates superior lifelong learning performance over existing methods, characterized by two strengths: reduced forgetting on previously learned tasks, and more effective cross-task transfer to unseen domains. Further analysis indicates that the task streaming affects performance, where consolidating structurally similar tasks facilitates agent efficacy.

Our contributions can be summarized as follows:

- We propose **LifeMem**, a novel lifelong learning framework for experience reuse across het-

erogeneous environments. By clustering trajectories into structured skills, LifeMem mitigates catastrophic forgetting while enabling cross-task transfer to unseen domains.

- To address the lack of agent-environment interaction trajectories in existing 4 environments, we contribute over 2k newly annotated trajectories as training data to enable agent learning.
- Through experimental analysis, we find that task streaming order impacts learning outcomes, and consolidating structurally similar trajectories within memory benefits agent performance.

2 Related Work

2.1 Lifelong Learning in LLMs

Lifelong learning enables LLMs to learn continuously and adaptively over their operational lifetime by integrating new knowledge while retaining previously acquired information, thereby mitigating catastrophic forgetting (Zheng et al., 2025b). (1) work on LLM lifelong learning primarily focuses on continual pretraining (Cossu et al., 2024; Gupta et al., 2023), fine-tuning (Sun et al., 2019; Huang et al., 2021), and alignment (Lin et al., 2024), which are conducted on a sequence of predefined static datasets. (2) work on LLM agents lifelong learning mainly investigate experience accumulation within a single environment (Wang et al., 2024; Zhao et al., 2024a), in which agents improve performance through repeated interactions under fixed state and action spaces.

Different from prior work, we focus on lifelong experience accumulation for agents across multiple environments (Zheng et al., 2025c). In this setting, agents continuously integrate interaction trajectories originating from heterogeneous environments and learn to transfer experience across domains while preserving previously encountered ones.

2.2 Memory Mechanism of LLM Agents

LLM agents rely on memory mechanisms to manage and reuse past experience because of the limited context window (Yu et al., 2025; Chhikara et al., 2025). To this end, researchers adopt external vector database that store interaction trajectories and retrieve relevant ones based on semantic similarity to support in-context learning (Packer et al., 2023; Liu et al., 2023a; Wang et al.; Zheng

et al., 2024). However, they fail to transfer experience to tasks with different environments or other out-of-distribution settings (Zhang et al., 2025).

More recent works explore abstract memory representations to guide agent planning. However, frameworks utilizing dialogue summaries or knowledge graphs (Chhikara et al., 2025; Rasmussen et al., 2025) primarily target conversation compression rather than workflow reuse. Though advanced methods like AutoSkill (Yang et al., 2026) and Hermes¹ attempt to distill high-level skills directly from past trajectories, their global extraction over an unorganized experience pool allows irrelevant trajectories to act as cross-environment noise, severely degrading the representativeness of each skill. In contrast, our approach clusters trajectories into structural workflows before distillation, yielding highly representative skills.

3 Preliminaries

We consider a LLM agent interacting with an observable environment $\mathcal{E} = (S, A, \Omega, T)$, where S denotes the environment states, A the action space, Ω the observation space, and T the transition dynamics. At each time step t , the agent receives an observation $o_t \in \Omega$ and selects an action

$$a_t = P_\theta(g, M, o_t) \quad (1)$$

where P_θ is the language model policy, g is a natural language goal, and M is memory. The interaction continues until task termination.

A task is defined as $\mathcal{T}^i = \langle \mathcal{E}^i, o_0^i, g^i \rangle$, where $\mathcal{E}^i$ denotes the environment of the i -th task, o_0^i is the initial observation, and g^i is the task goal. Solving a task produces a trajectory $\xi^i = \{(o_0, a_0), (o_1, a_1), \dots, (o_T, a_T)\}$.

In lifelong learning, the agent encounters a sequence of tasks $\{\mathcal{T}^1, \mathcal{T}^2, \dots, \mathcal{T}^N\}$. In the inter-environment setting, tasks arise from distinct environments $\mathcal{E}$. Trajectories collected from past tasks are incorporated into memory, as

$$M^i = M^{i-1} \cup \{\xi^i\} \quad (2)$$

to guide future task solving. The objective is to leverage accumulated experience to improve performance on new tasks while mitigating forgetting of previously learned behaviors.

4 Method

To enable experience accumulation and reuse in inter-environment lifelong learning, we model memory in an LLM agent as a dynamic component that is initialized once, incrementally updated with new experience, and queried during inference.

4.1 Memory Initialization

We initialize memory as a set of workflow-level clusters, which constitute the basic units for experience storage. Each cluster corresponds to a distinct behavioral workflow abstracted from trajectories.

Given an initial set of trajectories $\{\xi^0, \dots, \xi^N\}$, LLM partitions them into a set of clusters $C = \{c_1, \dots, c_M\}$ by identifying common workflow patterns. For each cluster $c_i \in C$, a concise description d_i is generated from its constituent trajectories, forming the initial memory.

4.2 Memory Evolution

Workflow Clustering: Since trajectories from different environments $\mathcal{E}$ may differ in goal semantics and action space while sharing underlying workflows, organizing them solely by semantic similarity is insufficient. We therefore update memory by assigning new trajectories to memory based on functional workflow alignment.

When a new trajectory ξ arrives, we retrieve its top- k related trajectories from memory and collect their associated clusters C_ξ . Conditioned on ξ and C_ξ , language model determines one of two actions: (1) if ξ aligns with an existing cluster $c \in C_\xi$, it's added to c and the corresponding description d_c is updated. (2) if ξ exhibits misalignment with C_ξ , a new cluster is created to accommodate it.

As experience accumulates, a workflow cluster may begin to encompass heterogeneous behaviors, thereby reducing specificity. To mitigate this issue, we introduce a geometric diversity constraint based on the Mean Resultant Length (MRL) from directional statistics (Mardia and Jupp, 2000). Given a cluster c containing n trajectories, we first normalize each trajectory's embedding representation to a unit vector $\mathbf{x}_i$ where $\|\mathbf{x}_i\| = 1$. The directional concentration R is defined as:

$$R = \left| \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \right| \quad (3)$$

The value $R \in [0, 1]$ directly quantifies structural homogeneity; $R \rightarrow 1$ implies high redundancy

¹<https://github.com/NousResearch/hermes-agent>

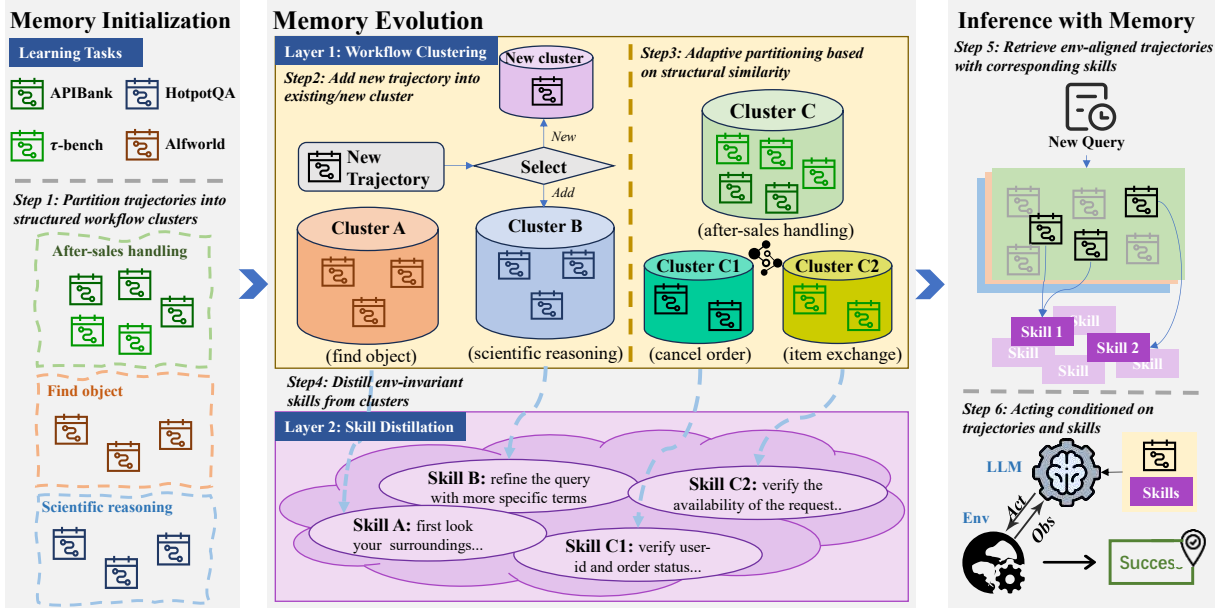


Figure 2: Overview of the LifeMem framework. In *memory initialization*, LifeMem groups interaction trajectories from heterogeneous environments into structured workflow clusters. In *memory evolution*, incoming trajectories are aligned with existing workflows or initialize a new one; as experience accumulates, workflow clusters are adaptively partitioned into finer-grained ones, from which transferable skills are distilled. In *inference*, LifeMem activates environment-aligned trajectories along with their skills to guide the agent’s actions.

(e.g., repeating identical templates), while a lower R indicates diverse operational behaviors. Accordingly, c is partitioned into finer-grained sub-clusters if and only if both the volume threshold and diversity constraint are satisfied: $n > L$ and $R < \tau_{mrl}$. This enables LifeMem to adaptively adjust workflow granularity based on trajectory similarity, ensuring partitioning occurs only under genuine behavior divergence.

Skill Distillation: Storing experience solely at the trajectory level limits continuous transfer, as trajectories collected in one environment seldom generalize to others due to mismatched observations or action spaces. To enable robust workflow reuse across heterogeneous tasks, we introduce a high-level skill abstraction layer, which distills each structured workflow cluster into concise, environment-invariant textual skills to guide subsequent agent execution.

Formally, for each workflow cluster $c_i \in C$, we map it to a corresponding high-level skill:

$$s_i = f(c_i), \quad (4)$$

where $f(\cdot)$ denotes the language model prompting process used to abstract trajectories contained in c_i . Each distilled skill s_i captures the shared constraints, logical dependencies, and generalized execution principles decoupled from low-level envi-

ronment details. The resulting hierarchical memory is formally denoted as:

$$\mathcal{H} = \{(c_i, s_i) \mid c_i \in C\}. \quad (5)$$

4.3 Inference with Memory

During inference, the agent must leverage past experience to solve the current task while avoiding interference from environment-specific behaviors. To this end, a dual-tier mechanism activates both environment-specific trajectories and environment-invariant skills to guide the agent.

Given a target task, we first retrieve the top- k most similar historical trajectories $\{\xi^0, \xi^1, \dots, \xi^{k-1}\}$ from the memory. These trajectories are conditioned on the same environment $\mathcal{E}$ to provide explicit, environment-aligned operational exemplars for task execution.

In parallel, we identify the workflow clusters C_ξ associated with these retrieved trajectories and activate their corresponding skills:

$$S_\xi = \{s_i \mid c_i \in C_\xi\} \quad (6)$$

Since trajectories from diverse environments sharing similar execution logic are grouped into the same clusters, this skill layer facilitates robust cross-environment transfer without introducing low-level action interference.

Category	Datasets	Trainset	Testset
Embodied Action	AlfWorld (Shridhar et al.)	2204	134
	ScienceWorld* (Wang et al., 2022)	2043	139
Tool Utilization	APIBank (Li et al., 2023)	3131	262
	τ -bench-airline (Yao et al., 2024)	530	115
Web Search	HotpotQA (Yang et al., 2018)	800	100
	Webshop (Yao et al., 2022a)	1571	100
Data Analysis	LifelongAgentBench (Zheng et al., 2025a)	280	-
	ToolQA-coffee (Zhuang et al., 2023)	700	80
Web Browser	Miniwob++ (Liu et al., 2018)	221	48
	Mind2web (Deng et al., 2023)	1009	252
Sum.		12489	1230

* We perform stratified sampling from ScienceWorld to ensure the proportions of different task topics.

Table 1: Overview of tasks and datasets used in our lifelong learning setup.

Ultimately, the agent generates actions conditioned on both the concrete exemplars and the high-level skills:

$$a = P_{\theta}(g, M_{act}, o) \quad (7)$$

where $M_{act} = (\{\xi^j\}_{j=0}^{k-1}, S_{\xi})$ denotes the activated memory contexts. This design allows LifeMem to benefit simultaneously from precise behavioral grounding and transferable skill abstractions.

5 Experiment

5.1 Dataset Construction

Overview: To evaluate the ability of agents to reuse experience across heterogeneous environments, we conduct experiments on 5 widely used agent scenarios following prior work (Chang et al., 2024; Liu et al., 2023b): embodied action (Shridhar et al.), tool utilization (Li et al., 2023), web search (Yang et al., 2018), data analysis (Zheng et al., 2025a), and web browsing (Liu et al., 2018). Specifically, we select ten tasks from diverse environments, comprising 12,489 training instances and 1,230 test instances (Table 1). Details of datasets and environments are in Appendix A.

Data generation: Since τ -bench (Yao et al., 2024), HotpotQA (Yang et al., 2018), Lifelon-

gAgentBench (Zheng et al., 2025a), and ToolQA (Zhuang et al., 2023) don’t provide training splits, we construct training data by collecting agent–environment interaction trajectories on them. Specifically, we prompt *GPT-4.1*² to solve these tasks under ReAct framework (Yao et al., 2022b), yielding 2,310 trajectories, among which 1,541 are successful according to environment feedback.

Quality control: We conduct human evaluation to assess the quality of generated trajectories (Appendix B). The results show that over 90% of the model’s actions are judged to be reasonable, indicating that the generated trajectories are of high quality and suitable for guiding model learning.

5.2 Baselines

We consider three baselines that represent different experience reuse strategies. ReAct (Yao et al., 2022b) serves as a **memory-free** baseline, which relies solely on step-by-step reasoning and action without past experience. In contrast, we adopt three representative **external-memory-based methods**: Synapse (Zheng et al., 2024) abstracts the current state and retrieves relevant trajectories for in-context learning, whereas ExpeL (Zhao et al., 2024a) and AutoSkill (Yang et al., 2026) shift towards extracting high-level experience from raw trajectories. Specifically, the former distills insights from trial and error, whereas the latter uncovers operational skills from successful rollouts. We use GPT-4.1 to extract experience for all baselines.

We use three widely adopted models for experiments: GPT-4o-mini (Achiam et al., 2023), Deepseek-v3.2-exp (Liu et al., 2025), and Qwen3-32b (Yang et al., 2025). They come from different families, span both proprietary and open-source settings, and vary in parameter scale, providing a representative evaluation across LLM backbones.

5.3 Evaluation Metrics

Following prior work (Srinivasan et al., 2022; Tomilin et al., 2025), the evaluation of lifelong learning can be viewed from two stages: the extent to which previously learned tasks transfer experience to the current task, and the impact of subsequently learned tasks to the current. Let $s_{1,\dots,j}^i$ denote the performance of task i after trained on tasks $1, \dots, j$, and let the number of training tasks be N , we assess the lifelong learning capability from following perspectives.

²<https://developers.openai.com/api/docs/models/gpt-4.1>

Model	Task	Overall Performance $\uparrow$					Backward Transfer $\uparrow$			
		ReAct	Synapse	ExpeL	AutoSkill	LifeMem	Synapse	ExpeL	AutoSkill	LifeMem
<i>GPT-4o-mini</i>	Embodied Action	10.62	40.02	44.09	12.36	45.65	-3.85	+8.81	-28.97	+14.84
	Tool Utilization	35.18	30.93	36.87	34.40	37.93	-11.33	-1.63	+2.84	+0.42
	Web Search	32.50	31.50	33.50	29.50	39.50	-9.89	+4.80	-6.65	+6.91
	Data Analysis	73.53	68.75	65.00	45.00	68.75	-3.51	+0.53	-2.70	-1.79
	Web Browsing	14.73	39.49	41.37	50.72	41.15	-2.70	+1.36	0.00	+2.77
	Avg.	28.84	39.18	41.85	33.22	44.13 $\uparrow$ 5.45%	-7.04	+3.10	-8.54	+5.66
<i>Deepseek-v3.2-exp</i>	Embodied Action	38.98	52.89	35.28	39.27	54.95	-3.17	+3.29	-17.03	-1.82
	Tool Utilization	49.34	53.69	54.07	52.87	52.70	-0.64	-1.10	+5.10	-1.56
	Web Search	30.00	36.00	39.50	27.00	40.00	-1.25	0.00	-26.86	+12.66
	Data Analysis	37.50	63.75	71.25	61.25	68.75	+2.00	-3.39	0.00	+1.85
	Web Browsing	48.07	34.28	32.84	33.04	34.08	+3.33	-1.61	+2.43	+3.33
	Avg.	41.14	46.38	43.85	40.62	48.02 $\uparrow$ 3.54%	-0.60	-0.28	-9.09	+3.11
<i>Qwen3-32b</i>	Embodied Action	28.09	52.72	55.62	35.41	55.61	+1.65	+3.52	+7.11	+9.27
	Tool Utilization	39.04	41.12	39.90	28.32	44.09	-3.29	-9.96	-8.81	+4.54
	Web Search	35.00	33.50	35.00	33.50	42.00	-2.57	+5.60	-8.07	0.00
	Data Analysis	37.50	67.50	75.00	47.50	70.00	-1.82	0.00	-28.36	+9.80
	Web Browsing	32.04	27.72	26.79	40.09	32.64	-7.70	-7.70	-2.69	+16.00
	Avg.	33.98	41.96	43.29	35.79	46.52 $\uparrow$ 7.46%	-2.24	-1.17	-6.17	+6.68

Table 2: Performance of various agent memory methods under inter-environment lifelong learning. For each category, the reported values are the averages over the datasets belonging to that category. **Avg.** denotes the average over all test datasets. The best and second-best results are marked **purple** and **orange**, respectively.

Overall Performance: We compute the average performance of the model across all tasks after learning all tasks (Tang et al., 2024):

$$OP = \frac{1}{N} \sum_{i=1}^N s_{1,\dots,N}^i \quad (8)$$

which provides an overall assessment perspective.

Backward Transfer: By comparing the model’s performance on task i after learning all tasks with that after learning only tasks $1, \dots, i$, we measure the impact of subsequent tasks on performance of task i (Tang et al., 2024; Srinivasan et al., 2022):

$$BwT = \frac{1}{N-1} \sum_{i=1}^{N-1} \frac{s_{1,\dots,N}^i - s_{1,\dots,i}^i}{s_{1,\dots,i}^i} \quad (9)$$

a positive value indicates that subsequent tasks may facilitate earlier tasks, whereas a negative value reflects forgetting caused by subsequent tasks.

We present these metrics in a more intuitive manner in Appendix C.

5.4 Implementation Details

In our memory implementation, we use SQLite³ to store all clusters and FAISS (Douze et al., 2024) to index and retrieve trajectories. Following prior

work (Lv et al., 2026), we utilize *all-MiniLM-L6-v2* (Wang et al., 2021a) as our embedding model. During the initialization stage, the number of initial trajectories is set to 5. In the memory evolution, the retriever returns the top-4 relevant trajectories, and the cluster splitting threshold is set to $L = 50$ and $\tau_{mrl} = 0.95$. We employ a lightweight language model *GPT-5-mini*⁴ to efficiently determine the cluster assignment of incoming trajectories. For cluster splitting and skill abstraction, we use a more powerful model *GPT-4.1*. Detailed prompts and hyper-parameters are in Appendix L.

Overall, we organize the lifelong task stream to reflect a common non-stationary deployment dynamic where temporally adjacent tasks often exhibit semantic and operational locality. Specifically, tasks from homogeneous environments are sequentially grouped to simulate distinct operational phases (AlfWorld $\rightarrow$ ScienceWorld $\rightarrow$ APIBank $\rightarrow$ τ -bench $\rightarrow$ HotpotQA $\rightarrow$ Webshop $\rightarrow$ LifelongAgentBench $\rightarrow$ ToolQA $\rightarrow$ MiniwoB++ $\rightarrow$ Mind2web). Within each category, tasks follow a coarse easy-to-hard progression to evaluate structural adaptation under structured data streams, adhering to standard curriculum learning benchmarks (Wang et al., 2021b). To further explore

³<https://sqlite.org/index.html>

⁴<https://developers.openai.com/api/docs/models/gpt-5-mini>

the system’s robustness, we analyze the impact of fully randomized and buffer-based streams in Section 6.3.

5.5 Main Results

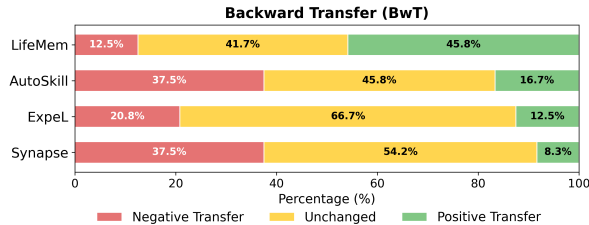


Figure 3: Distribution of transfer behaviors across all datasets during lifelong learning. Scores within the range of $[-5, 5]$ are categorized as "Unchanged" representing cases with no obvious performance shift.

Overall Performance: Table 2 reports the *overall performance* and *backward transfer* scores of different methods across five agent scenarios. LifeMem consistently outperforms all baselines across all metrics, demonstrating stronger overall lifelong learning performance. We further observe that experience-based learning yields larger performance gains for weaker backbone models. For example, GPT-4o-mini improves from 28.84 to 44.13, whereas the stronger model Deepseek-v3.2-exp shows a smaller increase from 41.14 to 48.02. This suggests that LifeMem benefits weaker models more substantially by distilling reusable skills from interaction trajectories to guide problem solving, thereby partially narrowing the performance gap across backbone models.

Transfer Behavior: To evaluate how subsequent learning impacts past knowledge, we analyze both the average BwT values and the distribution of transfer events (Figure 3). In baseline methods, as the number of learned tasks increases, the unstructured memory inevitably becomes cluttered. This leads to severe negative transfer due to inference interference: for the retrieval baseline Synapse, the cluttered memory causes the retrieval of irrelevant trajectories; the lack of experience isolation allows ongoing learning to overwrite and degrade the representativeness of early skills.

In contrast, LifeMem systematically structures the memory via explicit workflow clustering, effectively mitigating cross-task interference to achieve a superior, positive average BwT value in Table 2. By organizing trajectories into distinct, structurally-aligned clusters, LifeMem significantly suppresses negative transfer events and maximizes positive

transfer, allowing experience from subsequent tasks to consistently reinforce rather than interfere with earlier knowledge.

6 Analysis

In this section, we conduct a comprehensive analysis aimed at addressing the following research questions. **RQ1:** Which components of our method are critical to its performance in lifelong learning? (§6.1) **RQ2:** How effective is LifeMem in facilitating zero-shot cross-task experience transfer? (§6.2) **RQ3:** How do different task streaming strategies influence experience reuse performance? (§6.3) **RQ4:** Can boosting retrieval components directly bridge the performance gap between baselines and LifeMem? (§D) **RQ5:** What is the impact of hyperparameters? (§E)

6.1 RQ1: Ablation Study

Method	Overall Performance
ReAct (No memory)	
gpt-4o-mini	28.84
gpt-4.1	40.42
LifeMem	
+trajs	42.84
+skills (by gpt-4o-mini)	36.16
+skills (by gpt-4.1)	37.61
+trajs+skills (by gpt-4o-mini)	43.18
+trajs+skills (by gpt-4.1)	44.13

Table 3: Component-wise ablation study on LifeMem. We evaluate the incremental effects of our env-aligned trajectories (*trajs*) and abstracted skills (*skills*) across gpt-4o-mini as backbone. The model in parentheses indicates the generator used to extract the skills.

To dissect the contribution of each individual component within LifeMem, we conduct a multi-dimensional ablation study. Specifically, we evaluate the system performance under three configuration granularities: providing historical trajectories alone, incorporating abstracted workflows, and utilizing both components simultaneously.

Our ablation results in Table 3 reveal that both environment-aligned trajectories and abstracted skills consistently provide positive contributions to LifeMem. First, the inclusion of *trajs* provides raw demonstrations for the agent to mimic successful task execution formats, which is foundational for interactive agent tasks. Second, the abstracted *skills* further elevate the performance by offering macro-guidance, with the joint configuration yielding the optimal success rate.

Furthermore, we observe that the capacity of the skill generator plays a nuanced role. A more powerful gpt-4.1 exhibits a marginal advantage over gpt-4o-mini in extracting high-level skills from trajectories. Crucially, even when LifeMem relies entirely on gpt-4o-mini to extract skills, it still delivers a substantial gain over the baseline. Besides, LifeMem with a gpt-4o-mini backbone substantially outperforms the vanilla performance of gpt-4.1, verifying that the core efficacy of LifeMem stems from extracting and reusing skills from past trajectories, rather than distilling a superior LLM. We further present the ablation results of other models in Appendix F.

6.2 RQ2: Zero-Shot Cross-Task Transfer

Method	GPT-4o-mini	Deepseek-v3.2-exp
<i>FEVER</i>		
ReAct	0.40	0.39
Synapse	0.38	0.43
ExpeL	0.41	0.32
AutoSkill	0.49	0.36
LifeMem	0.49 _{↑22.5%}	0.46 _{↑17.9%}
LifeMem (w/o trajs)	0.47 _{↑17.5%}	0.43 _{↑10.3%}
<i>τ-bench (airline)</i>		
ReAct	0.32	0.38
Synapse	0.26	0.40
ExpeL	0.26	0.42
AutoSkill	0.34	0.36
LifeMem	0.38 _{↑18.8%}	0.44 _{↑15.8%}
LifeMem (w/o trajs)	0.38 _{↑18.8%}	0.42 _{↑10.5%}

Table 4: Performance of zero-shot experience transfer to unseen tasks. Percentages represent the cross-task transfer improvements compared to ReAct. *w/o trajs* means setting that only abstracted skills are provided.

To evaluate the zero-shot cross-task transfer capability of LifeMem, we investigate the following scenarios: (1) *transfer to unseen task formulations* within an encountered environment (evaluating on the fact verification task FEVER (Thorne et al., 2018) using the same Wikipedia environment from HotpotQA), and (2) *transfer to unseen environments* with a similar task formulation (evaluating on the airline task from τ -bench (Yao et al., 2024) against the retail task). These configurations rigorously test zero-shot transfer, forcing the agent to adapt accumulated experience to unfamiliar domains without prior task-specific training.

As shown in Table 4, baselines achieve limited gains or even suffer from performance degradation compared to ReAct. For the first setting (FEVER), the semantic shift in task formulation causes raw-retrieval methods (Synapse and ExpeL) to retrieve

irrelevant trajectories that misguide actions, while AutoSkill fails to match any compliant skill cluster for the unseen formulation. For the second setting, relying solely on raw trajectories fails to bridge the domain gap, as environmental discrepancies severely hinder direct experience reuse (See Appendix N for case study).

In contrast, LifeMem delivers strong zero-shot transfer by successfully leveraging environment-invariant skills that capture the domain-agnostic logic of task execution. Consequently, even when facing unseen task formulations or unfamiliar environments, LifeMem effectively transfers these high-level skills to guide agent actions, demonstrating superior flexibility in open-ended target domains.

6.3 RQ3: Impact of Task Stream Strategies

Task Stream	GPT-4o-mini	Deepseek-v3.2-exp
Similarity	44.13	48.02
Random	39.63	44.34
Buffer-based	41.64 _{↑44.7%}	46.73 _{↑64.9%}

Table 5: Robustness analysis of LifeMem across different task streaming strategies. Percentages denote the performance recovery rate relative to the degradation under the random stream.

Task stream sequence effects learning performance (Tang et al., 2024). We investigate three different streams: **similarity**, sequential learning of tasks from identical scenarios; **random**, which uses a completely shuffled instance sequence, and **buffer-based**, where incoming random trajectories are temporarily held in a minor task-specific cache (size=5) and committed as a consolidated mini-batch to stabilize the unstructured stream.

As reported in Table 5, grouping related tasks improves memory stability and reduces retrieval interference under streaming experience accumulation. Crucially, even when encountering uncurated real-world streams, LifeMem can be easily optimized via the buffer-based strategy, enabling seamless stabilization. More results in random settings are in Appendix H.

7 Conclusion

In this work, we propose LifeMem, a novel life-long learning framework that structures interaction trajectories into abstracted skills to enable effective experience reuse across heterogeneous environments. Evaluated on 10 environments and over

13k tasks, LifeMem significantly improves lifelong performance, successfully mitigating forgetting on learned tasks while achieving robust zero-shot cross-task transfer. Our analysis highlights a clear synergy: environment-aligned trajectories effectively guide action formats, whereas abstracted skills generalize better to unfamiliar scenarios. Ultimately, these insights provide practical design guidelines for how to better manage accumulated memory and extract reusable experience, enabling continual evolution after agent deployment.

Limitations

While our results establish the efficacy of LifeMem in integrating and reusing cross-environment experience during lifelong learning, fully unlocking its potential for large-scale industrial deployments introduces exciting opportunities for engineering-level co-optimization. For instance, transitioning this framework to real-world applications with millions of stored experiences could be further accelerated by deploying lightweight, on-device language models to ensure local efficiency, or incorporating approximate retrieval mechanisms alongside other system-level enhancements.

Ethical Consideration

All datasets used in this work are publicly available and released for research purposes. We strictly follow their original licenses and intended usage. For benchmarks without official training splits, we construct training data by collecting agent-environment interaction trajectories generated under controlled task-oriented environments. The collected data contains no Personally Identifiable Information (PII), sensitive content, or offensive material.

We ensure that all employed human annotators underwent rigorous training specific to the task categories involved in the evaluated environments. All participants were fully informed of the intended data usage prior to their involvement and were compensated fairly. Human evaluation is used solely to assess the reasonableness of model-generated interaction steps, ensuring data quality while minimizing subjective annotation.

Acknowledgments

We thank all the anonymous reviewers for their insightful and valuable comments. This work is

supported by the National Natural Science Foundation of China (Grant No.U21B2009).

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A Details of Datasets and Environments

A.1 Embodied Action

ALFWorld (Shridhar et al.) is an embodied household benchmark built on the ALFRED environment, where agents follow natural language instructions to perform multi-step tasks involving navigation, object manipulation and state changes (e.g., picking up, placing, opening) in simulated indoor scenes. The action space is discrete and text-based, supporting embodied reasoning and planning. We set the maximum episode length to 20 steps, and we report success rate as the evaluation metrics.

ScienceWorld (Wang et al., 2022) is an embodied interactive environment designed to evaluate scientific reasoning and procedural knowledge acquisition through language interaction. The environment simulates virtual laboratory and household settings, where agents follow textual instructions to complete tasks involving multi-step experiments, object manipulation, and causal reasoning (e.g., heating, mixing and measuring). Similar to ALFWorld, its action space is discrete and text-based, consisting of navigation, manipulation, and inquiry actions that allow agents to interact with objects and query the environment. We set its maximum step as 35, and report *Success Rate* as the evaluation metrics. Here we adopt a more stringent evaluation criterion: a task is considered successfully executed only when the environment returns a score of 100. This criterion helps better reflect the correctness of the model’s intermediate steps.

Both two datasets share similar embodied interaction settings with discrete, text-based action spaces, and ScienceWorld covers a wider range of task types about operations in science experiments and provides a more diverse set of actions.

A.2 Tool Utilization

APIBank (Li et al., 2023) is a tool-use benchmark that evaluates an agent’s ability to invoke external APIs to solve user-specified tasks (e.g., retrieving flight information, querying weather, or managing calendar events). Agent first uses ‘ToolSearcher’ to find suitable tools, call them with appropriate arguments and make next step based on calling results. It focuses on assessing tool utilization,

parameter reasoning, and decision making over multi-step API workflows. We select lv1 and lv2 level tasks as our testset, with maximum step as 5.

Retail (τ -bench) (Yao et al., 2024) evaluates agents in real-world vertical domains, including retail and airline services, with a focus on multi-step API tool usage. Agents receive natural language instructions and must execute sequences of tool calls, such as querying inventory, modifying reservations, or calculating prices, across multi-turn interactions with a database. We set its maximum step as 30.

Similar to APIBank, agent in τ -bench acts as assistant, interpreting user instructions and invoking a sequence of APIs to accomplish the requested tasks. This setup evaluates the agent’s ability to plan and execute multi-step operations in a coherent and goal-directed manner.

A.3 Web Search

HotpotQA (Yang et al., 2018) is a multi-hop question-answering dataset. Following prior approach (Zhao et al., 2024a), agent interacts with Wikipedia API to search for relevant information and generate answers. Its action space includes querying, navigating through pages, and producing answers, enabling evaluation of multi-step reasoning and information retrieval in a structured tool-use context. Following prior approach (Zhao et al., 2024a), *Exact Match* is used to measure the accuracy of the dataset, with maximum step as 7.

Webshop (Yao et al., 2022a) evaluates agents on multi-step online shopping tasks. It contains 1.18 million real-world products collected from Amazon. The environment is implemented using a ChromeDriver-driven browser, where simplified HTML pages serve as observations. Agents interact by entering search queries, clicking buttons, and navigating the page to find and purchase products that satisfy user requests. Once a purchase is finished, the environment provides a reward signaling the degree of match between the acquired product and the user’s specifications. *Task Score* is defined as the mean reward over all tasks, and *Success Rate* represents the fraction of instructions for which $r = 1$, signifying the complete fulfillment of all user criteria. We set its maximum step as 15.

Both two datasets evaluate the ability of agent to search for information on the web according to user requests, select the most relevant results, and make decisions based on the retrieved content.

A.4 Data Analysis

LifelongAgentBench (Zheng et al., 2025a): we select the “Database” task from LifelongAgentBench as a component of our training set for data analysis tasks. Within this environment, agents interact with a MySQL database via SQL statements to execute tasks. These tasks encompass 22 SQL-related skills, such as column aliasing, complex filtering with WHERE and HAVING clauses, multi-column grouping, data manipulation (INSERT, UPDATE, DELETE), and nested subqueries.

ToolQA (Zhuang et al., 2023) evaluates LLM’s ability to use external tools for question-answering. We select Tabular Database task in ToolQA, which requires LLM to perform real-time querying to an external database about coffee prices and compute through a calculator. ToolQA provides two modes of interaction with the database:

- *Custom functions* relies on author-defined functions such as LoadDB, GetValue and Filter, which utilize the pandas library for underlying tabular data queries.
- *SQLInterpreter* loads data into SQLite and executes queries by generating SQL statements.

Our training set comprises 210 trajectories based on custom functions and 259 based on SQLInterpreter, whereas the test set is entirely SQL-based. Agent is expected to extract common knowledge from the former to achieve superior experience generalization. We set its maximum step as 20.

Both datasets reflect the model’s capability to comprehend user requirements and interact with databases through interfaces such as SQL to retrieve relevant data for task completion.

A.5 Web Browser

Miniwob++ (Liu et al., 2018) is a web-based interaction benchmark that evaluates agents on a diverse set of tasks simulating basic human–computer interactions, such as clicking buttons, logging in, copying and pasting text, and booking flights. The environment presents agents with simplified web interfaces, where the state is represented by raw HTML code. Agents interact through a discrete action space including clicking elements, moving the mouse, typing text, and selecting options.

Mind2web (Deng et al., 2023) is a realistic benchmark for evaluating web agents on open-domain tasks across diverse real-world websites,

such as Airbnb and Twitter. It consists of human-annotated demonstrations that require agents to follow high-level user instructions and perform multi-step web navigation. The observation space is the raw HTML of the current webpage, while the action space includes primitive browser interactions such as click, type, and select. Agent performance is evaluated using task-level success rate, measuring whether all steps are correctly executed. We use the cross-task set in Mind2web as our dataset.

Both datasets evaluate an agent’s ability to accomplish diverse tasks through web browsing and interactive page manipulation. Compared to MiniWoB++, Mind2Web features more complex and realistic web environments and more challenging operations, closer to real-world web usage. The clear difference between the two benchmarks provides a natural curriculum, which is beneficial for studying curriculum learning and cross-environment knowledge transfer in web-based agents.

A.6 Out-of-Distribution Datasets

FEVER (Thorne et al., 2018) is a large-scale dataset for fact extraction and verification. As same as HotpotQA, agents in this environment interact with a Wikipedia-based search engine to retrieve evidence for verifying given claims. The task requires the agent to navigate through multiple documents, identify supporting or refuting information, and ultimately classify the claim as Supported, Refuted, or NotEnoughInfo. We evaluate the agent on a subset of 100 test instances, setting the maximum step as 4.

Airline (τ -bench) (Yao et al., 2024) evaluates agents in a complex aviation service context. Crucially, this environment presents a novel domain that was not encountered in our previous training tasks. In this domain, agent interacts with a simulated airline database to manage diverse user requests, such as searching for flights, rebooking itineraries, handling baggage claims, or calculating refund fees. Distinct from the retail task which focuses on inventory and order management, the airline domain introduces more stringent policy constraints and complex business logic, requiring the agent to handle time-sensitive resource allocation and multi-layered compensation rules. We evaluate the agents on a subset of 50 test cases, setting the maximum step as 30 for each interaction.

Trajectory Quality Evaluation

INTERACTION TRAJECTORY

USER

What was the opening price of coffee on 2007-09-27?

AGENT

We need to find the opening price of coffee on a specific date, so we should load the coffee database first. `LoadDB[coffee]`

ENVIRONMENT

We have successfully loaded the coffee database, including the following columns: Date, Open, High, Low, Close, Volume, Currency.

AGENT

We need to query the opening price for the date 2007-09-27 from the coffee database. `SQLInterpreter[SELECT Open FROM coffee_data WHERE Date = '2007-09-27']`

ENVIRONMENT

Open: 127.9

AGENT

The opening price of coffee on 2007-09-27 is 127.9. `Finish[127.9]`

ENVIRONMENT

Answer is CORRECT

REASONABLENESS ASSESSMENT

☐ 1. **Completely Correct:** The reasoning is logical and the steps are efficient.

☒ 2. **Generally Correct:** The goal was met, but some intermediate steps were redundant or unnecessary.

☐ 3. **Incorrect/Confused:** The agent's process is illogical, messy, or failed to follow the environment feedback.

Submit Evaluation

Figure 4: Screen shot of the annotation.

B Human Annotation in Dataset Construction

To ensure the quality of the newly generated correct trajectories, we conducted a rigorous human evaluation. We recruited five annotators, all of whom hold at least a Bachelor’s degree in computer science or related fields to ensure their technical proficiency in interpreting agent logs. All participants underwent rigorous training specifically tailored to the diverse task categories and environments involved. They were fully informed of the study’s objectives and intended data usage prior to their involvement. All annotators were compensated fairly at a rate of 1.5 times the local minimum wage.

The evaluation assesses the reasonableness of intermediate interaction steps, defined as whether each action represents a logical progression toward the task goal given the current observation. The results confirmed that over 90% of the intermediate steps in the generated trajectories were reasonable, validating the dataset’s suitability for guiding the lifelong learning of agents.

C An Intuitive Illustration of Evaluation Metrics

To facilitate understanding, we present our evaluation metrics: *Overall Performance* and *Backward Transfer*, in an intuitive manner.

Let $s_{1,\dots,j}^i$ denote the performance of task i after training on tasks $1, \dots, j$. As shown in Table 6, the rows below ALFWORLD correspond to 9 tasks learned sequentially. After completing training on each task j , we evaluate the model on all previously learned tasks $1, \dots, j$, resulting in a table with a lower-triangular structure. For example, after sequentially completed learning on ALFWORLD $\rightarrow$ SCIENCEWORLD, we evaluate the model on both datasets, achieving scores of 50.75 and 35.25, respectively (Row 2).

In addition, the SELF-LEARN row reports the performance s_i^i of the agent when trained and evaluated on a single task in isolation. For example, the second value from the left, 35.97, indicates the model’s performance on SCIENCEWORLD when it is trained exclusively on that task.

C.1 Overall Performance

Overall Performance denotes the average performance of the model across all tasks after it has completed learning on the entire task sequence; that is, the mean of all values highlighted in red in the table. This can be formally expressed as:

$$OP = \frac{1}{N} \sum_{i=1}^N s_{1,\dots,N}^i \quad (10)$$

C.2 Backward Transfer

Backward Transfer represents the impact of learning subsequent tasks $i, \dots, N$ to task i , which can be formally expressed as:

$$BwT = \frac{1}{N-1} \sum_{i=1}^{N-1} \frac{s_{1,\dots,N}^i - s_{1,\dots,i}^i}{s_{1,\dots,i}^i} \quad (11)$$

This can be intuitively seen by computing the difference between values after learning all tasks and values on corresponding dialogue in the same column. For example:

$$BwT_{miniweb} = (75.00 - 77.08) / 77.08 = -2.70\% \quad (12)$$

indicating that learning MIND2WEB has a negative effect on MINIWOB++.

Train/Eval	AlfWorld	ScienceWorld	APIBank	τ -bench	HotpotQA	Webshop	ToolQA	Miniwob++	Mind2web
AlfWorld	48.51								
ScienceWorld	50.75	35.25							
APIbank			45.42						
τ -bench	41.79	38.13	41.06	26.09					
HotpotQA					36.00				
Webshop	40.30	37.41	39.69	25.22	35.00	34.00			
ToolQA	45.52	34.53	35.88	23.48	31.00	32.00	71.25		
Miniwob++								77.08	
Mind2web	44.78	35.25	36.64	25.22	31.00	32.00	68.75	75.00	3.97

Table 6: Illustration of the lifelong learning process of GPT-4o-mini using the Synapse. SELF-LEARN denotes the performance of the model when trained and evaluated on a single dataset in isolation.

Configuration	Overall Performance
Synapse	39.18
Synapse (+ <i>advanced embedding</i>)	40.55
Synapse (+ <i>re-ranking</i>)	37.29
LifeMem	44.13

Table 7: Performance of Synapse equipped with advanced retrieval configuration.

D Investigation on Advanced Retrieval Components

To address **RQ4** and verify whether enhanced retrieval capabilities can fundamentally mitigate cross-environment interference, we implement two advanced retrieval variations on top of the Synapse baseline using the *GPT-4o-mini* backbone:

- **Advanced Embedding:** We substitute the default embedding retriever with a higher-tier model *text-embedding-ada-002*⁵, to evaluate the impact of enhanced semantic matching.
- **Semantic Retrieval + Re-ranking:** To further optimize candidate precision, we utilize *BGE-reranker-v2-m3* (Luo et al., 2024) to re-rank a broader pool of $5 \times k$ retrieved candidates, subsequently delivering the top- k highest-scoring trajectories to the agent.

As reported in Table 7, simply elevating the retrieval components yields marginal or even detrimental effects on agent execution. Upgrading to a stronger embedding model only provides a negligible performance gain, failing to closely approach the superiority of LifeMem. More noticeably, integrating re-ranking mechanism severely degrades the performance. This result may occurs because

⁵<https://developers.openai.com/api/docs/models/text-embedding-ada-002>

off-the-shelf, pre-trained re-ranking models are inherently optimized for query-document text relevance in informational retrieval, rather than the intricate behavioral and logic alignment required between sequential agentic trajectories.

These empirical insights firmly demonstrate that simply refining retrieval mechanisms is insufficient to mitigate cross-environment interference, thereby validating the architectural necessity of LifeMem’s hierarchical skill abstraction.

E Hyperparameter Sensitivity Analysis

To evaluate the robustness of LifeMem, we conduct hyper-parameter sensitivity experiments on the AlfWorld using *GPT-4o-mini*. We systematically investigate three key parameters: the retrieval size ($top - k$), the maximum cluster capacity (L), and the cluster split threshold coefficient (τ_{mrl}).

Hyperparameter	Value	Success Rate (%)
Retrieval Size ($top - k$)	2	56.72
	4	58.21
	8	58.21
Cluster Size Limit (L)	10	48.51
	30	52.99
	50	58.21
	80	51.49
Split Coefficient (τ_{mrl})	0.9	54.48
	0.95	58.21
	1.0	57.46

Table 8: Hyperparameter sensitivity analysis.

Impact of Retrieval Hyperparameter: $top - k$ controls the number of retrieved trajectories used for in-context learning. We vary $top - k \in \{2, 4, 8\}$ while keeping clustering parameters constant. Results are listed in Table 8.

The success rate remains stable, demonstrating that the system performance is relatively insensitive

to the retrieval volume, indicating that LifeMem relies more on high-quality workflow abstractions than instance quantities.

Impact of Workflow Clustering Parameters: the workflow clustering granularity is governed by the cluster size limit L and the split threshold coefficient τ_{mrl} (where $\tau_{mrl} = 1.0$ means splitting depends solely on L). We evaluate $L \in \{10, 30, 50, 80\}$ and $\tau_{mrl} \in \{0.9, 0.95, 1.0\}$, revealing the impact of clustering boundaries:

- **Under-clustering:** when L is too large or τ_{mrl} is too low, distinct trajectories merge into single bloated clusters. This under-clustering dilutes the representativeness of the abstracted skills, failing to provide precise guidance and thus degrading performance.
- **Over-clustering:** when L is too small, it may partition data into fragmented sub-clusters too aggressively. This causes LifeMem to overfit to circumstantial noise within specific rollouts rather than forming generalizable workflows, which limits transferability.

F Ablation Study

	GPT-4o-mini	Deepseek-v3.2-exp	Qwen3-32b
LifeMem	44.13	48.02	46.52
LifeMem (w/o skills)	42.84	47.09	43.88
LifeMem (w/o examples)	37.61	43.46	39.65

Table 9: Ablation study across three backbone models. *LifeMem (w/o skills)* evaluates performance without the abstracted skills, while *LifeMem (w/o examples)* removes env-aligned trajectories during inference.

G Costs in Learning and Inference Time

Method	Tokens	Characters
Synapse	1.88k	11.57k
LifeMem	2.18k	12.72k
ExpeL	2.21k	13.07k
AutoSkill	4.68k	28.73k

Table 10: Comparison of token consumption and length of prompt across methods.

In **memory construction**, LifeMem scales approximately linearly with the number of trajectories (nearly \$1 per 1k trajectories). By utilizing the

lightweight model, the overall cost remains moderate and sustainable for long-term lifelong settings.

At **inference time**, we measured the average context length of the initial agent query across all benchmarks, including the system prompt, few-shot exemplars, abstracted skills, and current task descriptions. As shown in Table 10, token consumption is dominated by system prompts and retrieved trajectories, while LifeMem adds only about 300 tokens over Synapse but yields a 12.63% relative performance improvement. Under the GPT-4o-mini pricing, this translates to less than \$1 additional inference cost across our whole test-set. In contrast, AutoSkill incurs a substantially larger prompt footprint because it lacks workflow-aware organization and therefore injects a bloated global skill profile containing many irrelevant cross-environment behaviors.

H Performance in Random Settings

	Synapse	ExpeL	AutoSkill	LifeMem
GPT-4o-mini	39.18	39.37	32.13	39.63
Deepseek-v3.2-exp	41.14	40.58	41.88	44.34

Table 11: Overall performance on random settings.

As shown in Table 11, we additionally evaluate LifeMem on the fully random task stream. Although the random task stream is more challenging, as it disrupts opportunities to reuse related experiences, LifeMem consistently remains the best-performing memory approach.

I Analysis of Memory Clusters

Cluster Purity. To examine whether LifeMem organizes trajectories according to consistent execution workflows, we conduct a human annotation study to evaluate cluster purity and analyze the statistical properties of the learned memory clusters. Specifically, three annotators examine the trajectories within each sampled cluster, identify the dominant macro-workflow, and mark trajectories whose execution logic deviates from it. We define Workflow Purity as the proportion of trajectories consistent with the dominant workflow:

$$Purity = \frac{\sum_{k=1}^K N_{consistent,k}}{N_{total}} \quad (13)$$

Across all sampled clusters, LifeMem achieves an average workflow purity of 91%, providing human evidence that the learned clusters capture

consistent execution workflows rather than merely grouping trajectories by embedding similarity.

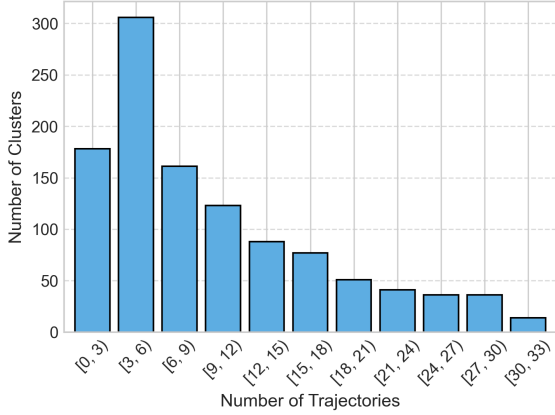


Figure 5: Distribution of the number of trajectories contained in different clusters under the **similarity** learning order.

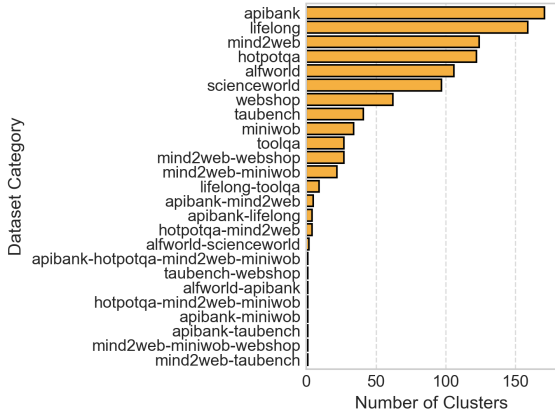


Figure 6: Number of cluster types across datasets, categorized by the datasets to which trajectories within each cluster belong.

Adaptive Cluster Granularity. We further analyze the statistical properties of the learned clusters in Figure 5 and Figure 6. As shown in Figure 5, most clusters consolidate multiple trajectories, indicating that LifeMem can abstract recurring execution patterns across tasks instead of storing isolated case-specific experiences.

The cluster statistics also reveal that the learned granularity adapts to the diversity of execution patterns. For environments with relatively repetitive workflows, such as WebShop, clusters contain 29 trajectories on average. In contrast, for more diverse environments such as τ -bench, the average cluster size decreases to 12 trajectories. Consistently, Figure 6 shows that datasets with

more diverse semantic and action patterns, such as APIBank, are partitioned into a larger number of specialized cluster types, whereas more homogeneous environments, such as AlfWorld and ToolQA, result in fewer and broader clusters. Together, these results suggest that LifeMem can adapt its memory granularity to the structural diversity of trajectories, avoiding both excessive fragmentation and the merging of heterogeneous workflows without manual specification.

J Detailed Numerical Results

We present the results of different task training orders on each dataset in Tables 12.

We present detailed results of different methods on each dataset in Tables 13 of our main results.

K Examples of the Dataset

We present examples of the four datasets we constructed in Figures 7, 8, 9, 10.

L Details of Implementation

We set the temperature of all models to 0.3. For Deepseek-v3.2-exp and Qwen3-32B, we do not enable their reasoning modes.

We present the prompts used in building our agent memory system in Tables 14, 15, 16, and 17.

M Prompts for task execution

We present the prompts of each dataset used in inference time of LifeMem in Tables 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, and 28.

N Failure Case Study

Here, we present a representative failure case study on the FEVER dataset. Since the new task description differs from all previously learned tasks, Synapse may retrieve irrelevant trajectories (e.g., from APIBank). These trajectories exhibit substantial discrepancies in the action space compared to the FEVER environment, thereby misleading the agent’s action generation and ultimately causing task failure. In contrast, our method retrieves environment-aligned trajectories and restricts retrieval to the corresponding HotpotQA repository, effectively avoiding this issue. Moreover, we further provide abstracted insights that offer reusable experience for the model.

Dataset	GPT-4o-mini			Deepseek-v3.2-exp		
	Similarity	Random	Buffer-based	Similarity	Random	Buffer-based
AlfWorld	58.21	44.03	52.24	55.22	42.54	58.21
SciWorld	33.09	23.74	23.02	54.68	55.40	51.80
APIbank	45.42	44.27	43.89	39.31	43.89	45.42
Tau-bench	30.43	27.83	28.70	66.09	63.48	60.87
HotpotQA	42.00	41.00	40.00	40.00	32.00	32.00
Webshop	37.00	31.00	35.00	40.00	39.00	40.00
ToolQA	68.75	65.00	70.00	68.75	56.25	63.75
Miniwob++	78.72	76.60	78.72	64.58	62.50	64.58
Mind2web	3.57	3.17	3.17	3.57	3.97	3.97
Avg.	44.13	39.63	41.64	48.02	44.34	46.73

Table 12: Performance of GPT-4o-mini and Deepseek-v3.2-exp with LifeMem under Different Learning Orders.

Model	Dataset	Overall Performance					Backward Transfer			
		ReAct	Synapse	ExpeL	AutoSkill	LifeMem	Synapse	ExpeL	AutoSkill	LifeMem
GPT-4o-mini	AlfWorld	10.45	44.78	51.49	6.72	58.21	-7.69	-6.75	-66.65	+14.70
	SciWorld	10.79	35.25	36.69	17.99	33.09	0.00	+24.37	+8.70	+14.98
	APIbank	44.27	36.64	41.60	46.18	45.42	-19.33	-6.03	+1.67	+6.25
	Tau-bench	26.09	25.22	32.17	22.61	30.43	-3.33	+2.78	+4.00	-5.41
	HotpotQA	38.00	31.00	32.00	23.00	42.00	-13.89	+6.67	-25.81	+5.00
	Webshop	27.00	32.00	35.00	36.00	37.00	-5.88	+2.94	+12.50	+8.82
	ToolQA	73.53	68.75	65.00	45.00	68.75	-3.51	-1.89	-2.70	-1.79
	Miniwob++	27.08	75.00	79.17	97.87	78.72	-2.70	+2.71	0.00	+2.77
	Mind2web	2.38	3.97	3.57	3.57	3.57	-	-	-	-
	Avg.	28.84	39.18	41.85	33.22	44.13	-7.04	+3.10	-8.54	+5.66
Deepseek-v3.2-exp	AlfWorld	47.01	60.45	21.64	43.28	55.22	-7.95	+3.54	-30.13	-6.34
	SciWorld	30.94	45.32	48.92	35.25	54.68	+1.61	+3.03	-3.92	+2.70
	APIbank	43.89	43.89	44.66	43.13	39.31	+2.67	+1.75	+2.74	-7.22
	Tau-bench	54.78	63.48	63.48	62.61	66.09	-3.95	-3.95	+7.47	+4.11
	HotpotQA	33.00	33.00	42.00	31.00	40.00	0.00	0.00	-27.91	+21.21
	Webshop	27.00	39.00	37.00	23.00	40.00	-2.50	0.00	-25.81	+5.26
	ToolQA	37.50	63.75	71.25	61.25	68.75	+2.00	-3.39	0.00	+1.85
	Miniwob++	93.75	64.58	62.50	62.50	64.58	+3.33	-3.22	+4.87	+3.33
	Mind2web	2.38	3.97	3.17	3.57	3.57	-	-	-	-
	Avg.	41.14	46.38	43.85	40.62	48.02	-0.60	-0.28	-9.09	+3.11
Qwen3-32b	AlfWorld	41.79	70.90	72.39	48.51	71.64	+9.19	+3.19	+12.08	-1.04
	SciWorld	14.39	34.53	38.85	22.30	39.57	-5.89	+3.85	+3.34	+19.58
	APIbank	45.04	37.02	38.93	30.54	41.22	-14.92	-0.97	-5.86	+9.08
	Tau-bench	33.04	45.22	40.87	26.09	46.96	+8.34	-18.96	-11.77	0.00
	HotpotQA	39.00	37.00	38.00	30.00	46.00	-5.13	-7.32	-18.92	0.00
	Webshop	31.00	30.00	32.00	37.00	38.00	0.00	+18.52	+2.78	0.00
	ToolQA	37.50	67.50	75.00	47.50	70.00	-1.82	0.00	-28.36	+9.80
	Miniwob++	61.70	51.06	50.00	76.60	61.70	-7.70	-7.70	-2.69	+16.00
	Mind2web	2.38	4.37	3.57	3.57	3.57	-	-	-	-
	Avg.	33.98	41.96	43.29	35.79	46.52	-2.24	-1.17	-6.17	+6.68

Table 13: Detailed results of each dataset on the performance of different models under inter-environment lifelong learning.

Prompt for Categorizing Trajectories into Specific Skill Cluster

Role & Mission

You are an advanced reasoning agent responsible for organizing task trajectories into skill-based clusters. You are operating in incremental assignment mode. Your role is to assign new trajectories to existing clusters or to create a new cluster if necessary. You must NOT delete, merge, or split existing clusters in this mode.

Each trajectory corresponds to a task execution, and tasks may require overlapping or distinct underlying skills (e.g., planning, tool use, symbolic reasoning, perception, long-horizon decomposition).

Input Data

New Trajectories:

{trajectories}

You also have access to an existing memory cluster library. Below are the EXISTING CLUSTERS retrieved as the most relevant candidates for comparison. This is a partial view of the memory, not an exhaustive list of all clusters. Each cluster represents a group of trajectories that rely on similar core skills:

{existing_clusters}

Your goal is to group trajectories according to the similarity of the skills required to solve the tasks, focusing on what capabilities are exercised rather than superficial task details or domains. Solve the problem with the following steps:

Step 1: Analysis (Chain of Thought)

Before outputting the final operations, perform a systematic analysis for each trajectory.

Deconstruct Skills: Identify the core reasoning requirements of the trajectory.

Evaluate Overlap: Compare these skills against the existing_clusters.

Determine Operation:

ADD: If the trajectory's skills overlap significantly or fit within the scope of an existing cluster. Prioritize broadening an existing cluster over creating a new one.

NEW: Only if the required skills represent a completely different domain with no meaningful overlap.

Step 2: Final Output (Operations)

Based on your analysis, output the actions using these operations:

ADD <TRAJECTORY_ID> to <CLUSTER_NAME>

NEW <CLUSTER_NAME> with <TRAJECTORY_ID>

Response Format:

Please format your response as follows:

<Analysis> [Provide a brief, step-by-step reasoning process here. Identify the 3–6 sub-skills found in the trajectories and explain why certain IDs belong together.]

<Execution>

NEW <Cluster_A> with <ID_1>

ADD <ID_3> to <Cluster_A>

Guidelines & Constraints:

Format: <CLUSTER_NAME> must use underscores (_) instead of spaces. TRAJECTORY_ID must match the 'id' field.

Assignment: A trajectory may belong to only one cluster. You can only add one trajectory per operation line.

Skill Focus: Base decisions on core skills and reasoning requirements, not task topics. Do not invent unrelated skills.

Cluster Creation: Create a NEW cluster only when a clear and defensible skill gap exists. Descriptions should focus on reasoning type and primary capabilities.

Table 14: Prompt for categorizing trajectories into specific skill cluster.

Prompt for Cluster Splitting

Role

You are an advanced reasoning agent specialized in deconstructing and refining skill-based memory clusters. You operate in **CLUSTER SPLIT** mode to ensure high-precision skill categorization.

Context

A memory cluster has grown too large (>30 trajectories), indicating it now covers too many distinct skills. You must dissolve the old cluster and redistribute **ALL** trajectories into 3–6 new, more granular clusters.

– Old Cluster Name:

{cluster_id}

– Old Cluster Summary:

{summary}

Input: Trajectories to Redistribute

{examples}

Task Instructions

1. **Deconstruction Analysis (CoT):** First, analyze the provided trajectories. Identify the underlying granular reasoning patterns (e.g., 'Temporal_Logic_Reasoning', 'Nested_Tool_Coordination'). Group them mentally based on structural similarity rather than broad topics.
2. **Assignment:** Assign EVERY trajectory ID from the input list to exactly one new cluster.

Table 15: Prompt for cluster splitting.

Prompt for Getting Cluster Summary

You are an advanced reasoning agent responsible for organizing task trajectories into skill-based clusters. Here are a few trajectories in one cluster, your task is to summarize these trajectories into one sentence, which is a description of the cluster, like their shared core skills, or the core tasks they can solve.

{trajectories}

Previous Summary:

{summary}

Output format:

Your summary should begin with "This cluster is about ..."

Table 16: Prompt for getting cluster summary.

Prompt for Abstracting skills

You are an advanced reasoning agent that can generate a concise rule set based on forming new insights from past task trajectories. You will be given successful trials.

Successful trials:

{success_history}

By examining the successful trials, you will extract 3 to 5 rules that are general and high-level insights of the successful trajectories or proposed Way of Thought. These rules should serve as helpful tips for different tasks in the future. Have an emphasis on tips that help the agent perform better Thought and Action.

Follow the below format:

RULE 1: rule content...

RULE 2: rule content...

Do not mention the trials in the rules because all the rules should be GENERALLY APPLICABLE. Each rule should be concise and easy to follow. Generate at least 3 and at most 5 rules in total.

Table 17: Prompt for abstracting skills.

Prompt for AlfWorld

You are alfred. Follow the syntax of the examples closely when taking actions. RULE: The command to place an object is ALWAYS 'move <object> to <location>'. Do not use any other format. You may take maximum of 20 steps.

Available actions:

look: look around your current location

inventory: check your current inventory

go to RECEP: move to a receptacle

take OBJ from RECEP: take an object from a receptacle

move OBJ to RECEP: move an object in or on a receptacle

examine OBJ/RECEP: examine an object or receptacle

use OBJ: use an object

open RECEP: open a receptacle

close RECEP: close a receptacle

clean OBJ with RECEP: clean an object using a receptacle

heat OBJ with RECEP: heat an object using a receptacle

cool OBJ with RECEP: cool an object using a receptacle

Here are two examples:

Examples:

{EXAMPLES}

(END OF EXAMPLES)

Here are some skill docs extracted from past experience, that may help you to complete this task:

Skills:

{SKILLS}

Now it's your turn!

{TASK}

Table 18: Prompt for AlfWorld dataset.

Prompt for ScienceWorld

You are an agent in ScienceWorld. Follow the syntax of the examples closely when taking actions.

RULE: The command to place an object is ALWAYS 'move <object> to <location>'. Do not use any other format. You may take maximum of 35 steps.

Available actions:

open/close OBJ: open/close a container
de/activate OBJ: activate/deactivate a device
connect OBJ to OBJ: connect electrical components
disconnect OBJ: disconnect electrical components
use OBJ [on OBJ]: use a device/item
look around: describe the current room
look at OBJ: describe an object in detail
look in OBJ: describe a container's contents
read OBJ: read a note or book
move OBJ to OBJ: move an object to a container
pick up OBJ: move an object to the inventory
put down OBJ: drop an inventory item
pour OBJ into OBJ: pour a liquid into a container
dunk OBJ into OBJ: dunk a container into a liquid
mix OBJ: chemically mix a container
go to LOC: move to a new location
teleport to LOC*: teleport to a specific room
eat OBJ: eat a food
flush OBJ: flush a food
focus on OBJ: signal intent on a task object
wait [DURATION]: take no action for some duration
task: describe current task
inventory: list agent's inventory

Here are some examples:

Examples:

{EXAMPLES}

Please note that these examples may only be similar to your current task, as their action space may differ, or they may hold limited relevance to the task at hand.

Here are your operational guidelines from skills extracted by past experiences:

Operational guidelines:

{SKILLS}

Current state and task:

{TASK}

Table 19: Prompt for ScienceWorld dataset.

Prompt for APIBank

You are an AutoGPT, capable of utilizing numerous tool-functions to complete the given task. I will provide you the task description <Task> with tools in a library. As there are many tools in library, you have to use the tool 'ToolSearcher' first to find some relevant tools you may needed. After that, you can use the retrieved tools to complete the task.

You should notice that:

1. The tool-calling format is: [tool_name1(parm1=value1, parm2=value2...)].
2. When finish the task, use the tool 'Finish' to end this task.
3. Some tasks may require you to use multiple tools to complete, which means you may call another tool based on the output of the previous tool.
4. When error occurred in the tool's output, you should modify the tool-calling format to make it correct.
5. If the tool has executed but returned nothing, you need to determine whether:\n(a) the tool truly has no return value , or (b) there was a problem during its use (such as an incorrect tool name or invalid parameters).

Tool descriptions:

"name": "ToolSearcher", "description": "Search for the relevant tools", "input_parameters": {"keywords": {"type": "str", "description": "The keywords used for search."}}, "output_parameters": {"token": {"type": "str", "description": "The description of the retrieved tool."}}
"name": "Finish", "description": "call this tool when you think the task is finished"

Here are two examples:

Example:

{EXAMPLE}

Here are some skills extracted from past experience, that may help you to complete this task:

Insights:

{INSIGHTS}

Task:

{TASK}

Table 20: Prompt for APIBank dataset.

Prompt for τ -Bench (Retail)

Retail Agent Policy

{POLICY}

Here are some skill docs extracted from past experience, that may help you to complete this task:

Skills:

{SKILLS}

Here are some examples:

Examples:

{EXAMPLES}

Task:

{TASK}

Table 21: Prompt for τ -Bench (Retail) dataset. Please refer to Yao et al. (2024) for the description of the agent policy.

Prompt for τ -Bench (Airline)

Prompt for Airline (tau-Bench)

Airline Agent Policy
{POLICY}

Here are some skill docs extracted from past experience, that may help you to complete this task:

Skills:
{SKILLS}

Examples:
{EXAMPLES}

Task:
{TASK}

Table 22: Prompt for τ -Bench (Airline) dataset. Please refer to [Yao et al. \(2024\)](#) for the description of the agent policy.

Prompt for FEVER

You are fact verifier. Solve a question answering task with interleaving Thought, Action, Observation steps. Thought can reason about the current situation, and Action can be three types:

- (1) Search[entity], which searches the exact entity on Wikipedia and returns the first paragraph if it exists. If not, it will return some similar entities to search.
- (2) Lookup[keyword], which returns the next sentence containing keyword in the last passage successfully found by Search.
- (3) Finish[answer], which returns the answer and finishes the task, answer should be one of REFUTES, SUPPORTS or NOT ENOUGH INFO.

You may take maximum of 4 steps.

Examples:
{EXAMPLES}

Here are some skill docs extracted from past experience, that may help you to complete this task:

Skills:
It is MANDATORY for you to follow these skills as CLOSELY as possible as they will help you perform the fact verification tasks efficiently:
{SKILLS}

Now it's your turn!

Claim:
{TASK}

Table 23: Prompt for FEVER dataset.

Prompt for HotpotQA

You are QA system. Solve a question answering task with interleaving Thought, Action, Observation steps. Thought can reason about the current situation, and Action can be three types:

- (1) Search[entity], which searches the exact entity on Wikipedia and returns the first paragraph if it exists. If not, it will return some similar entities to search.
- (2) Lookup[keyword], which returns the next sentence containing keyword in the last passage successfully found by Search.
- (3) Finish[answer], which returns the answer and finishes the task.

You may take maximum of 7 steps.

Examples:

{EXAMPLES}
(END OF EXAMPLES)

Here are some skill docs extracted from past experience, that may help you to complete this task:

Skills:

{SKILLS}

Now it's your turn!

{TASK}

Table 24: Prompt for HotpotQA dataset.

Prompt for Webshop

You are webshop. Follow the syntax of the examples closely when taking actions.

You may take maximum of 15 steps.

Here are two examples:

Examples:

{EXAMPLES}
(END OF EXAMPLES)

Insights:

The following are some skills you gathered on tasks of purchasing items requested by an user by interacting with an online website. Use these experiences as helpful references to help you perform better on this task:

{SKILLS}

Now it's your turn!

{TASK}

Table 25: Prompt for Webshop dataset.

Prompt for ToolQA

Solve a question answering task with interleaving Thought, Action, Observation steps. Thought can reason about the current situation, and Action can be 13 types:

- (1) Calculate[formula], which calculates the formula and returns the result.
- (2) LoadDB[DBName], which loads the database DBName and returns the database. The DBName is coffee.
- (3) SQLInterpreter[SQL], which interprets the SQL query SQL and returns the result.
- (4) Finish[answer], which returns the answer and finishes the task.

You may take as many steps as necessary. Use SQL query only. No other tools are allowed.

Here are some skills extracted from past experience, that may help you to complete this task:

Skills:

{SKILLS}

Here are two examples:

Examples:

{EXAMPLES}

(END OF EXAMPLES)

The prefix of line type is added automatically, do not output it again. One thought, one action and do not pretend to output observation.

You may get no results after you filtered database, you can use LoadDB[DBName] to load database again.

The numerical results are rounded to one decimal place.

Call the tool strictly following the format.

Question:

{TASK}

Table 26: Prompt for ToolQA dataset.

Prompt for Miniwob++

You are a large language model trained to navigate the web. To accomplish the task, use methods in the following Agent class to generate actions until you need the new state to proceed.

```
```class Agent:
def __init__(self, args):
Action: type a string via the keyboard
def type(self, characters: str) -> None:
Action: click an HTML element with a valid xpath
def click_xpath(self, xpath: str):
Actions: press a key on the keyboard, including:\n
enter, space, arrowleft, arrowright, backspace, arrowup, arrowdown, command+a, command+c, command+v
def press(self, key_type: str) -> None:
Action: click an option HTML element in a list with a valid xpath
def click_option(self, xpath: str):
Action: move mouse cursor on an HTML element with a valid xpath
def movemouse(self, xpath: str):
```
```

Here are some skill docs extracted from past experience, which may help you to complete the task:

Skills:

{SKILLS}

Examples:

{EXAMPLES}

Task:

{TASK}

Table 27: Prompt for Miniwob++ dataset.

Tau-bench

USER

My name is Omar Anderson and my zip code is 19031. I'm logical, independent, relaxing, polite. Return #W6067464 via credit_card_4190576: Electric Kettle; Wall Clock

AGENT

```
{\"name\": \"find_user_id_by_name_zip\", \"arguments\":  
  {\"first_name\": \"Omar\", \"last_name\": \"Anderson\", \"zip\": \"19031\"}}
```

USER

API output: omar_anderson_3203\", {\"content\": \"Action:\\n{\\n\"name\\\": \\\"get_order_details\\\",
\\arguments\\\": {\\\"order_id\\\": \\\"#W6067464\\\"}}

AGENT

```
{\"name\": \"get_order_details\", \"arguments\": {\"order_id\": \"#W6067464\"}}
```

USER

API output: {\\\"order_id\\\": \\\"#W6067464\\\", \\\"user_id\\\": \\\"omar_anderson_3203\\\", \\\"address\\\":
{\\\"address1\\\": \\\"845 Willow Lane\\\", \\\"address2\\\": \\\"Suite 906\\\", \\\"city\\\": \\\"Philadelphia\\\", \\\"country\\\":
\\\"USA\\\", \\\"state\\\": \\\"PA\\\", \\\"zip\\\": \\\"19031\\\"}...

AGENT

```
{\"name\": \"return_delivered_order_items\", \"arguments\":  
  {\"order_id\": \"#W6067464\", \"item_ids\":  
    [\"9624127908\", \"8917609800\"], \"payment_method_id\": \"credit_card_4190576\"}}
```

USER

API output: {\\\"order_id\\\": \\\"#W6067464\\\", \\\"user_id\\\": \\\"omar_anderson_3203\\\", \\\"address\\\":
{\\\"address1\\\": \\\"845 Willow Lane\\\", \\\"address2\\\": \\\"Suite 906\\\", \\\"city\\\": \\\"Philadelphia\\\", \\\"country\\\":
\\\"USA\\\", \\\"state\\\": \\\"PA\\\", \\\"zip\\\": \\\"19031\\\"}...

AGENT

I've requested the return for the following items from your order #W6067464...I've finished the task.
Here is the result: The return has been successfully requested.

USER

Thank you for your help! I appreciate it.

Figure 7: An example of taubench.

ToolQA (Coffee Database)

USER

What was the opening price of coffee on 2007-09-27?

AGENT

We need to find the opening price of coffee on a specific date, so we should load the coffee database first.

```
LoadDB[coffee]
```

USER

We have successfully loaded the coffee database, including the following columns: Date, Open, High, Low, Close, Volume, Currency.

AGENT

We need to query the opening price for the date 2007-09-27 from the coffee database.

```
SQLInterpreter[SELECT Open FROM coffee_data WHERE Date = '2007-09-27';]
```

USER

Open: 127.9

AGENT

The opening price of coffee on 2007-09-27 is 127.9.

```
Finish[127.9]
```

USER

Answer is Correct.

Figure 8: An example of ToolQA.

LifelongAgentBench (Database)

USER

Insert a new payment record for member ID 102 with a payment date of '2023-10-15', amount 75, and payment method 'Credit Card' into the membership payments table. The name of this table is membership_payments, and the headers of this table are payment_id, member_id, payment_date, amount, payment_method.

AGENT

To insert a new payment record into the `membership\_payments` table, we need to use the `INSERT INTO` SQL statement. The values we need to insert are: \n\n- `member\_id`: 102\n- `payment\_date`: '2023-10-15'\n- `amount`: 75\n- `payment\_method`: 'Credit Card'\n\nThe `payment\_id` is typically an auto-incrementing primary key, so we do not need to specify it in the insert statement. \n\nThe SQL statement will look like this: \n\n```\nINSERT INTO membership_payments (member_id, payment_date, amount, payment_method) VALUES (102, '2023-10-15', 75, 'Credit Card');\n```\n\nNow, I will execute this SQL statement to insert the new payment record.

```
Action: Operation
```sql
INSERT INTO membership_payments (member_id, payment_date, amount,
payment_method) VALUES (102, '2023-10-15', 75, 'Credit Card');```
```

USER

Executed successfully.

AGENT

The SQL statement executed successfully, indicating that the new payment record for member ID 102 has been inserted into the `membership\_payments` table. Since the operation was successful and there are no further actions required for this task, I will now provide the final answer.

```
Action: Answer
Final Answer: []
```

USER

Answer is correct.

Figure 9: An example of LifelongAgentBench.

## HotpotQA

USER

Firststar Center was the name for two buildings before the company changed its name to match which company based in Minneapolis?

AGENT

*I need to find out which company Firststar changed its name to, and that company is based in Minneapolis. I will start by searching Firststar Center.*

Search[Firststar Center]

USER

Heritage Bank Center is an indoor arena in downtown Cincinnati, adjacent to Great American Ball Park. It was completed in September 1975 and named Riverfront Coliseum because of its placement next to Riverfront Stadium. In 1997, the facility became known as The Crown, and in 1999, it changed its name again to Firststar Center after Firststar Bank assumed naming rights  
...[wikipedia information]...

AGENT

*The passage says that after Firststar's merger with U.S. Bank, the arena was renamed U.S. Bank Arena. I need to confirm if U.S. Bank is based in Minneapolis.*

Search[U.S. Bank]

USER

U.S. Bancorp (stylized as us bancorp) is an American multinational financial services firm headquartered in Minneapolis, Minnesota and incorporated in Delaware. It is the 5th-largest bank in the United States as of 2025. As the largest bank in the Midwestern United States, it is considered systemically important by the Financial Stability Board. It is the parent company of its primary operating entity, U.S. Bank National Association, which does business as U.S. Bank (stylized as us bank)  
...[wikipedia information]...

AGENT

*U.S. Bancorp (U.S. Bank) is headquartered in Minneapolis, Minnesota. After Firststar's merger with U.S. Bank, the name was changed to U.S. Bank Arena. Therefore, the company based in Minneapolis that Firststar changed its name to match is U.S. Bank.*

Finish[U.S. Bank]

USER

Answer is Correct.

Figure 10: An example of HotpotQA.

Prompt for Mind2web

You are a large language model trained to navigate the web. Output the next action and wait for the next observation.

- Action space:**
- 1. `CLICK [id]`: Click on an HTML element with its id.
  - 2. `TYPE [id] [value]`: Type a string into the element with the id.
  - 3. `SELECT [id] [value]`: Select a value for an HTML element by its id.

Here are some skill docs extracted from past experience, which may help you to complete the task:

**Skills:**  
{SKILLS}

**Examples:**  
{EXAMPLES}

**Task:**  
{TASK}

Table 28: Prompt for Mind2web dataset.

## Failure Case

### # Task Description

**Claim:** The heart beats at a resting rate close to 22 beats per minute.

You are a fact verifier. Solve a question answering task with interleaving **Thought**, **Action**, **Observation** steps. **Thought** can reason about the current situation, and **Action** can be three types: **Search[entity]**: searches the exact entity on Wikipedia and returns the first paragraph if it exists. If not, it will return some similar entities to search.

**Lookup[keyword]**: returns the next sentence containing the keyword in the last passage successfully found by **Search**.

**Finish[answer]**: returns the answer and finishes the task. The answer should be one of **REFUTES**, **SUPPORTS**, or **NOT ENOUGH INFO**.

You may take a maximum of 4 steps.

### # Irrelevant Examples Retrieved from APIBank

**User:** I want to add a new exercise to the list for the chest muscle group. The name of the new exercise should be "Chest Fly," and the description should be "Lie down on a bench with your feet flat on the ground and hold a dumbbell in each hand. Extend your arms straight up, then slowly lower them out to the side until your elbows are at shoulder level, and bring them back up."

**Assistant:** [ToolSearcher(keywords='add a new exercise to the list')]

**User:** "API: add\_exercise | Description: Add a new exercise to the list of available exercises ..."

**Assistant:** [add\_exercise(name='Chest Fly', description='Lie down on a bench with your feet flat on the ground and hold a dumbbell in each hand. Extend your arms straight up, then slowly lower them out to the side until your elbows are at shoulder level, and bring them back up.', muscle\_group='chest')]

**User:** {"data": {"status": "success"}}

**Assistant:** [Finish()]

### # Bad Inference

**Claim:** The heart beats at a resting rate close to 22 beats per minute.

**Action:** [Search[heart rate]]

**Observation:** Invalid Action. Valid Actions are Lookup[<topic>] Search[<topic>] and Finish[<answer>].

**Action:** [Search[resting heart rate]]

**Observation:** Invalid Action. Valid Actions are Lookup[<topic>] Search[<topic>] and Finish[<answer>].

Table 29: Failure Case

## Success Case

### # Task Description

**Claim:** The heart beats at a resting rate close to 22 beats per minute.

You are a fact verifier. Solve a question answering task with interleaving **Thought**, **Action**, **Observation** steps. **Thought** can reason about the current situation, and **Action** can be three types: **Search[entity]**: searches the exact entity on Wikipedia and returns the first paragraph if it exists. If not, it will return some similar entities to search.

**Lookup[keyword]**: returns the next sentence containing the keyword in the last passage successfully found by **Search**.

**Finish[answer]**: returns the answer and finishes the task. The answer should be one of **REFUTES**, **SUPPORTS**, or **NOT ENOUGH INFO**.

You may take a maximum of 4 steps.

### # Env-align Examples Retrieved from HotpotQA

**User:** Swiss music duo Double released their best known single "The Captain of Her Heart" in what year?

**Assistant:** Search[Double (music duo)]

**User:** Could not find [Double (music duo)]. Similar: ['Double (band)', ...]

**Assistant:** Search[Double (band)]

**User:** Double was a Swiss music duo best known for their hit single "The Captain of Her Heart".

**Assistant:** Lookup[The Captain of Her Heart]

**User:** The duo was formed in 1983 in Zürich ...

**Assistant:** Finish[1986]

### # Insights

The following paragraph are insights a teacher agent provided to you. It is **MANDATORY** for you to follow these insights as **CLOSELY** as possible as they will help you perform the fact verification tasks efficiently:

In order to successfully complete factual verification tasks, begin by clearly understanding the claim ...

### # Success Inference

**Claim:** The heart beats at a resting rate close to 22 beats per minute.

**Action:** Search[resting heart rate]

**Observation:** Heart rate is the frequency of the heartbeat ...

**Action:** Search[bradycardia definition]

**Observation:** Arrhythmias, also known as cardiac arrhythmias, are irregularities ...

**Action:** Finish[REFUTES]

**Observation:** Answer is CORRECT

Table 30: Success Case